

Sessional 1 (B.Tech VI Sem CSE)
BCE-C625 Theory of Computation: Classical to Quantum Models

Time: 1HR

Max. Marks :20

Note: Attempt all questions.

Questions	Marks	Learning Outcome	Bloom's Taxonomy															
Q1 Define the term Finite Automata. Differentiate between NFA and DFA.	4	CO1	U															
Q2. Write regular expression for the language over the alphabet {0,1}. With no consecutive zeors.	4	CO1	U															
Q3. Design a DFA equivalent for the given NFA; where q3 is the final state. <table border="1"><tr><td>Current State</td><td>Input 'a'</td><td>Input 'b'</td></tr><tr><td>→ q0</td><td>q0, q1</td><td>q0 , q2</td></tr><tr><td>q1</td><td>--</td><td>q3</td></tr><tr><td>q2</td><td>q0, q3</td><td>q1</td></tr><tr><td>q3</td><td>q2</td><td>-</td></tr></table>	Current State	Input 'a'	Input 'b'	→ q0	q0, q1	q0 , q2	q1	--	q3	q2	q0, q3	q1	q3	q2	-	8	CO3	E
Current State	Input 'a'	Input 'b'																
→ q0	q0, q1	q0 , q2																
q1	--	q3																
q2	q0, q3	q1																
q3	q2	-																
Q4. Design a FA for (0*1*) +	4	CO2	K															

A- Analyze , U- Understand, AP - Apply, E-Evaluate, K-Knowledge

Sessional 2 (B.Tech VI Sem CSE)
BCE-C625 Theory of Computation: Classical to Quantum Models
Time: 1HR

Date : 22.04.2026

Max. Marks :20

Note: Attempt all questions.

Questions	Marks	Learning Outcome	Bloom's Taxonomy
Q1 Define PDA.	4	CO1	U
Q2. What is an ambiguous grammar.	4	CO1	U
Q3. For the given grammar do left most derivation for the string aaabbbb ; Also draw the parse tree. $S \rightarrow ab \mid ba \mid aB \mid bA$; $A \rightarrow a \mid bAA$; $B \rightarrow b \mid aBB$	8	CO3	E
Q4. Design a PDA made of a's and b's which accepts only palindrome.	4	CO2	K

A- Analyze , U- Understand, AP - Apply, E-Evaluate, K-Knowledge